Our paper studies contact between lithium and the solid electrolyte inside a battery. If small gaps appear between them, current has a harder time passing through. We look at how pressure and the contact present at the start of discharge affect this resistance. This connects to nanomaterials because thin coatings and tiny defects can affect contact at the same boundary.

Phase-field simulation is a computer method for showing how the shape of a material changes over time. It can help us study how lithium builds up unevenly or how a small gap grows while a battery is used. Pressure and ion movement can also be included.

For our nanomaterials program, a phase-field model could let us compare different coating properties, pore sizes and tiny defects before deciding what to make and test. Our paper gives us a battery contact problem that this method could explore in more detail. The purpose is to guide experiments toward materials that keep better contact and let current pass more easily.